Edgardo Villar-Sepúlveda

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Curriculum Vitae

Research profile

My research lies at the intersection of nonlinear dynamics, applied mathematics, and mathematical biology, with a focus on pattern formation in nonlinear partial differential equations. I study Turing and Turing–wave instabilities, bifurcations, nonlinear waves, localised patterns, and homoclinic snaking in reaction–diffusion–transport systems. My recent work combines asymptotic analysis and dynamical systems, with a particular focus on Maxwell points and beyond-all-order asymptotics. Beyond this core research area, I have also supervised MSc projects in other fields such as financial technology and data science.

Positions held

2026–Present **TUM Global Postdoc Fellow**, *School of Computation, Information and Technology*, Technical University of Munich, Germany.

2024–2026 **Teaching Associate**, *School of Engineering Mathematics and Technology*, University of Bristol, UK.

2021–2024 **Teaching Assistant**, *School of Computer Science, Electrical and Electronic Engineering and Engineering Mathematics*, University of Bristol, UK.

2020 **Engineering lecturer**, *Facultad de Ingeniería y Ciencias*, Universidad Adolfo Ibáñez, Viña del Mar, Chile.

2018–2020 **Engineering lecturer**, *Departamento de Matemática*, Universidad Técnica Federico Santa María, Valparaíso, Chile.

Education

2021–2025 **PhD in Engineering Mathematics**, *University of Bristol*, UK.
Supervisor: Prof. Alan Champneys. **Date of award:** 17 June 2025. **Thesis:** General approaches to pattern formation in reaction–diffusion systems.

2018–2019 **MSc in Mathematics**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile.
Supervisor: Dr. Pablo Aguirre Olea.

2014–2019 **BSc in Mathematical Engineering**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile, with Academic Distinction.

Teaching training

2026 **ProLehre Onboarding**, *Technical University of Munich (completed October 2026).*
Elective module: Supervising Students' Theses (16 work units), covering structured supervision meetings, constructive feedback at each stage of a thesis, and criterion-based marking. Work units are recognised towards the advanced level of the Certificate for Teaching in Higher Education of the Bavarian Universities.

Fellowships and Awards

2026–Present **TUM Global Postdoc Fellowship**, *Technical University of Munich*, Host: Prof. Christian Kühn, ∼ € 75,000.
Awarded through a competitive international selection process to support independent postdoctoral research at TUM.

2025 **Patrick Woods Prize**, *University of Bristol*.
Best PhD Thesis of the School of Engineering Mathematics and Technology for the academic year 2024–2025. Monetary award: £2,500.

2024 **Student Travel Award**, *SIAM*.
Funded to attend the SIAM Conference on Nonlinear Waves and Coherent Structures in Baltimore, Maryland, USA. US$ 975.

2023 **Student Travel Award**, *SIAM*.
Funded to attend the SIAM Conference on Applications of Dynamical Systems in Portland, Oregon, USA. US$ 950.

2021 **Bristol PLUS Award**, *University of Bristol*.
Awarded to fewer than 5% of students annually for exceptional extracurricular engagement, including over 90 hours of teaching assistance and activities supporting professional and personal development.

2020 **Beca Chile de Doctorado en el Extranjero**, *Agencia Nacional de Investigación y Desarrollo*, Santiago, Chile.
Scholarship awarded through a competitive merit-based selection process to fund my PhD and living expenses. ∼ US$ 200,000.

2019 **Academic Merit Award**, *Universidad Técnica Federico Santa María*, Valparaíso.
Awarded to the top student among all those who entered the university in 2014, across all programmes. ∼ US$ 5,000.

2014–2019 **Dean's Honor List**, *Universidad Técnica Federico Santa María*, Valparaíso.
Ranked among the best students at my university each year. In 2018 and 2019, I ranked second among all enrolled students, across all entry-year cohorts.

2013 **Beca Puntaje Nacional**, *Universidad Técnica Federico Santa María*, Valparaíso.
Scholarship awarded to fund my undergraduate and master's studies for obtaining a perfect score in the University entrance test in Mathematics. ∼ US$ 35,000.

Publications

Published Journal Articles

[13] **Edgardo Villar-Sepúlveda**. Beyond-all-order asymptotics for homoclinic snaking of localised patterns in reaction-transport systems. *Nonlinearity*, 39(9):095025, 2026.

[12] **Edgardo Villar-Sepúlveda**, Alan R Champneys, Davide Cusseddu, and Anotida Madzvamuse. Pattern formation of bulk-surface reaction-diffusion systems in a ball. *SIAM Journal on Applied Mathematics*, 86(1):21–51, 2026.

[11] Andrew L Krause, Václav Klika, **Edgardo Villar-Sepúlveda**, Alan R Champneys, and Eamonn A Gaffney. Pattern localization in the Swift–Hohenberg equation via slowly varying spatial heterogeneity. *SIAM Journal on Applied Dynamical Systems*, 24(4):2804–2847, 2025.

[10] Fasma Diele, Andrew L Krause, Deborah Lacitignola, Carmela Marangi, Angela Monti, and **Edgardo Villar-Sepúlveda**. Transient instability and patterns of reactivity in diffusive-chemotaxis soil carbon dynamics. *Bulletin of Mathematical Biology*, 87(11):162, 2025.

[9] **Edgardo Villar-Sepúlveda**, Alan R Champneys, and Andrew L Krause. Designing reaction-cross-diffusion systems with Turing and wave instabilities. *Journal of Mathematical Biology*, 91(4):37, 2025.

[8] Molly Brennan, Andrew L Krause, **Edgardo Villar-Sepúlveda**, and Christopher B Prior. Pattern formation as a resilience mechanism in cancer immunotherapy. *Bulletin of Mathematical Biology*, 87(8):106, 2025.

[7] Chunyi Gai, **Edgardo Villar-Sepúlveda**, Alan Champneys, and Michael J Ward. An asymptotic analysis of spike self-replication and spike nucleation of reaction-diffusion patterns on growing 1-D domains. *Bulletin of Mathematical Biology*, 87(4):48, 2025.

[6] **Edgardo Villar-Sepúlveda** and Alan Champneys. Amplitude equations for wave bifurcations in reaction–diffusion systems. *Nonlinearity*, 37(8):085012, 2024.

[5] Alfredo E Calderón and **Edgardo Villar-Sepúlveda**. Piecewise contracting maps on the interval: Hausdorff dimension, entropy, and attractors. *Canadian Mathematical Bulletin*, 67(2):318–327, 2024.

[4] **Edgardo Villar-Sepúlveda** and Alan Champneys. Computation of Turing bifurcation normal form for n-component reaction-diffusion systems. *ACM Transactions on Mathematical Software*, 49(4):1–24, 2023.

[3] **Edgardo Villar-Sepúlveda** and Alan Champneys. Degenerate Turing bifurcation and the birth of localized patterns in activator-inhibitor systems. *SIAM Journal on Applied Dynamical Systems*, 22(3):1673–1709, 2023.

[2] **Edgardo Villar-Sepúlveda**, Pablo Aguirre, and Víctor F Breña-Medina. A case study of multiple wave solutions in a reaction-diffusion system using invariant manifolds and global bifurcations. *SIAM Journal on Applied Dynamical Systems*, 22(2):918–950, 2023.

[1] **Edgardo Villar-Sepúlveda** and Alan R Champneys. General conditions for Turing and wave instabilities in reaction-diffusion systems. *Journal of Mathematical Biology*, 86(3):39, 2023.

Submitted Preprints

[P1] Alfredo E Calderón and **Edgardo Villar-Sepúlveda**. Unifying topological entropy notions for piecewise continuous maps on the interval. *arXiv:2404.10867*, 2024.

Teaching

2024–2026 **Teaching Associate, University of Bristol**.

MSc level: I was the Unit Director for the MSc course Mathematical Modelling in Biology, Medicine and Public Health ($\sim$ 15 students), where I designed lectures and assessments and coordinated the teaching team. I also taught Introduction to Artificial Intelligence and Data Analytics ($\sim$ 60 students); Visual Analytics ($\sim$ 300 students); and contributed to Software Development: Programming and Algorithms ($\sim$ 40 students); Statistical Computing and Empirical Methods; Large-Scale Data Engineering for Business.

Undergraduate level: I taught Engineering Mathematics 1 and 2, courses serving $\sim$1,050 and $\sim$850 students, respectively; and contributed to Data Engineering at Scale; Data Science and Applied Statistics.

I also served as a personal tutor for undergraduate and master's students and supervised Financial Technology master's projects.

2021–2024 **Teaching assistant, University of Bristol**.

MSc level: I worked as a Teaching Assistant for Data Science Mini Project (∼100 students) throughout this period, serving as lead Teaching Assistant in 2022. In this master's pre-thesis course, I supervised group research projects proposed by industry partners. I also assisted with Software Development: Programming and Algorithms (∼40 students); Visual Analytics (∼300 students).
Undergraduate level: I also assisted with Data Science and Applied Statistics (∼ 60 students); Introduction to Data Science; Perspectives in Mathematics; Mathematical and Data Modelling; Introduction to Computer Programming; Final Year Project; Nonlinear Dynamics and Chaos (∼ 60 students); Numerical Methods in MATLAB (∼ 60 students); Applied Linear Algebra; Engineering Mathematics 1 and 2; Discrete Mathematics.
Furthermore, I gained experience supporting the Advanced Mathematics Support Programme's Problem Solving Matters course, which aims to develop Year 12 students' mathematical thinking and problem-solving skills.

2020 **Engineering teacher, Universidad Adolfo Ibáñez**.

I taught Linear Algebra and Multi-Variable Integral Calculus (∼ 30 students).

2018–2020 **Universidad Técnica Federico Santa María: Engineering teacher**.

I taught Algebra and Geometry (∼ 40 students); and Single- and Multi-Variable Differential and Integral Calculus (∼ 40 students).

2014–2017 **Teaching assistant, Universidad Técnica Federico Santa María**.

I assisted with Differential Geometry of Curves and Surfaces (∼ 20 students); Single and Multi-Variable Differential and Integral calculus (∼ 80 students); and Linear Algebra (∼ 40 students).

Conferences, Seminars, and Study Groups

Invited Talks

July 2026 **Dynamics Days Europe**, *Rectorate of Universidade NOVA de Lisboa and Colégio Almada Negreiros*, Lisbon, Portugal, Minisymposium: Singularly perturbed multiscale systems.
Maxwell points, the origin of nontrivial connections in multiscale systems

May 2026 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *Concordia University*, Montréal, Canada, Minisymposium: Coherent structures in biology and social sciences.
Maxwell points: The origin of homoclinic snaking and localised patterns beyond all asymptotic orders

May 2026 **Applied Nonlinear Dynamics seminar**, *University of Leeds*, UK.
Maxwell points, the origin of nontrivial connections in multiscale systems

May 2026 **Oberseminar Dynamics seminar**, *Technical University of Munich*, Germany.
Maxwell points, the origin of nontrivial connections in multiscale systems

March 2026 **British Applied Mathematics Colloquium**, *University of East Anglia*, Norwich, UK, Minisymposium: The next generation: frontiers in nonlinear waves and pattern formation.
Wave instabilities: the second face of Turing

May 2025 **SIAM Conference on Applications of Dynamical Systems**, *Sheraton Denver Downtown Hotel*, Denver, Colorado, US, Minisymposium: Waves in continuous and discrete domains: theory and applications.
Exponential asymptotics for the study of 'localised spatiotemporal patterns' on the line

May 2025 **Dynamical Systems seminar**, *University of Hamburg, Germany*.
Homoclinic snaking & localised patterns beyond all asymptotic orders

February 2025 **Dynamical Systems seminar**, *Leiden University, The Netherlands*.
Homoclinic snaking & localised patterns beyond all asymptotic orders

June 2024 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *Lord Baltimore Hotel*, Baltimore, Maryland, US, Minisymposium: Reaction–diffusion mechanisms for localised patterns and waves.
Travelling and standing waves as a consequence of cross-diffusion

May 2023 **SIAM Conference on Applications of Dynamical Systems**, *DoubleTree by Hilton Hotel*, Portland, Oregon, US, Minisymposium: Pattern formation in nature: from Busse balloons to homoclinic snaking.
Spatial instabilities as a consequence of Turing or transcritical bifurcations in bulk–surface reaction–diffusion equations on a sphere

April 2023 **British Applied Mathematics Colloquium**, *University of the West of England*, Bristol, UK, Minisymposium: Dynamics of reaction–transport systems.
Homoclinic snaking & localised patterns beyond all asymptotic orders

November 2022 **Mathematics of beyond all-orders phenomena**, *Isaac Newton Institute for Mathematical Sciences*, Cambridge, UK, Workshop.
Homoclinic snaking and the emergence of localised patterns beyond all asymptotic orders

September 2022 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *University of Bremen*, Germany, Minisymposium: Pattern formation, cell polarity and morphogenesis.
Turing patterns in general n-component reaction–diffusion equations posed on different geometries

Organisation of Minisymposia and Seminars

May 2026 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *Concordia University*, Montréal, Canada.
Organised Minisymposium (separate from my invited talk above): *Pattern formation, homoclinic snaking and modulation equations in non-standard systems* with Sebastian Suckau and Lütfiye Masur

2022–2024 **Engineering Mathematics Research Group seminars**, *University of Bristol*.
I organised seminars at weekly meetings we held with the members of the Engineering Mathematics Research Group

Contributed Talks

June 2025 **British Mathematical Colloquium & British Applied Mathematics Colloquium**, *University of Exeter*, UK.
Homoclinic snaking & localised patterns beyond all asymptotic orders

February 2025 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada, Online.
Exponential asymptotics to study the extent to which localised patterns exist

April 2024 **British Applied Mathematics Colloquium**, *University of Newcastle*, UK.
Turing and wave instabilities in reaction–diffusion systems with cross-diffusion

November 2023 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada, Online.
Diffusion-driven instabilities in reaction–diffusion systems with cross-diffusion

May 2023 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada.
Spatial instabilities as a consequence of Turing or transcritical bifurcations in bulk–surface reaction–diffusion equations on a sphere

January 2023 **Mathematical Biology seminar**, *The University of British Columbia*, Vancouver, Canada, Online.
Spatial and spatiotemporal oscillations in reaction–diffusion equations

June 2022 **Early Career Researchers in Mathematics**, *University College London*, London, UK.
Turing patterns in bulk–surface reaction–diffusion equations

April 2022 **British Applied Mathematics Colloquium**, *University of Loughborough*, UK, Online.
Turing instabilities in bulk–surface reaction–diffusion equations on a sphere

September 2021 **Celebrating Alan Turing's work in mathematical biology**, *University of Sheffield*, UK.
Necessary and sufficient conditions for an $n-$component system to show a Turing or Turing-Hopf instability pattern

January 2020 **World Conference on Natural Resource Modeling**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile.
Spatial diffusion in a predator-prey model with a strong Allee effect on the prey and a ratio-dependent functional response.

April 2019 **XXXII Jornada Matemática de la Zona Sur**, *Universidad de Magallanes*, Punta Arenas, Chile.
Ondas viajeras y caos en un modelo depredador-presa con respuesta funcional razón-dependiente y difusión espacial.

January 2019 **Tercer Workshop de Modelamiento Matemático de Sistemas Biológicos**, *Universidad Tecnológica Metropolitana*, Santiago, Chile.
Análisis de la influencia de la difusión espacial en un modelo depredador-presa.

Internal talks

May 2026 **Engineering Mathematics seminar**, *University of Bristol.*
Maxwell points, the origin of nontrivial connections in multiscale systems

April 2025 **Engineering Mathematics Postgraduate seminar**, *University of Bristol.*
Homoclinic snaking and localised patterns beyond all asymptotic orders

December 2022 **Engineering Mathematics seminar**, *University of Bristol.*
Homoclinic snaking & localised patterns beyond all asymptotic orders

June 2022 **PhD Away Day**, *University of Oxford*, Prize for the exceptional scientific content of my talk.
Pattern formation in reaction–diffusion equations

December 2021 **Engineering Mathematics Postgraduate seminar**, *University of Bristol.*
Necessary and sufficient conditions for a reaction–diffusion equation with n components to show a Turing or Turing-Hopf instability pattern

Study Groups

May 2024 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: The future of the high street.

January 2024 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Food security.

November 2023 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Maths for justice.

October 2022 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Recovery from the pandemic: Hospitality & leisure.

July 2022 **European Study Group with Industry**, *University of Kent*, Canterbury.
Theme: Alternate transforms for signal processing.

July 2021 **Virtual Knowledge Exchange in Mathematical Sciences**, Online.
Theme: Covid safety in large events.

MSc thesis supervision

Technical University of Munich: Mathematics

2026–Present **Barış Arslan**, *Exploratory numerical continuation of reaction–diffusion systems.*
Co-supervised with Christian Kühn.

University of Bristol: Financial Technology and Data Science

2025 **Tianyu Dong**, *Advanced modeling methods for stock market: time series prediction and multi-market exploratory clustering evaluation.*

2025 **Hanwen Li**, *Advanced methods to forecast the stock market: A unified benchmark of LSTM, CNN–LSTM and Attention–LSTM for long-horizon stock prediction.*

2025 **Dushyant Goel**, *Predictive modeling and risk analysis for UK start-up fundraising markets with an interactive decision-support tool for investors and founders.*

2025 **Biao Yin**, *Deep learning approaches for forecasting the S&P 500 Index: A comparative study of LSTM, CNN, and a hybrid model.*

2025 **Zhengbo Xu**, *UK tax incentives and startup financing dynamics: A multi-platform study of the early-stage ecosystem.*

2025 **Tangrong Li**, *Time-series forecasting of stock prices using transformer variants: A critical evaluation of Informer, Autoformer, and PatchTST.*

2025 **Li Yin**, *Research on the adaptability of advanced deep learning methods in stock prediction of different companies.*

University of Bristol: Engineering Mathematics

2022 **Weiyi Yuan**, *Mathematical modelling of infection spread within a relatively static crowd.* Co-supervised with Alan Champneys.

Academic and Administrative Service

Journal reviewing *Proceedings of the Royal Society A*; *Nonlinearity*; *Chaos, Solitons and Fractals*; *Physica D: Nonlinear Phenomena*; *IMA Journal of Applied Mathematics*; *Calculus of Variations and Partial Differential Equations*; *Journal of Nonlinear Science*; *Journal of Mathematical Biology*; *Computational and Applied Mathematics*; *Qualitative Theory of Dynamical Systems*; *Computational Methods for Differential Equations*; *Nonlinear Dynamics*; *Cognitive Computation*.

2024–2026 **Served as a personal tutor for undergraduate and MSc students**.
I supported students transitioning to University and their postgraduate studies.

2025 **Provided academic references for students applying for PhD positions**.

2025 **Second marker for three MSc dissertations in financial technology and data science**.

2021–2024 **Committee Member, Break-Away Bristol**.
I organised events and served as treasurer of Break-Away Bristol, one of the student societies at the University of Bristol.